

PROOF OF CONCEPT

Vitsika

Malagasy ant identification from a single photograph

vitsika is Malagasy for ant. Genus-level, 27 genera, built on public data and a frozen public model.

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Vitsika Malagasy ant genus identification · proof of concept

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QUERY IMAGE



casent0002219_p.jpg · embedded in 1078 ms

GENUS PREDICTION

Royidris 37%
subfamily myrmicinae

✓ Held-out test specimen casent0002219 (AntWeb label: *Royidris notorthotenes*) · correct

① Classifier confidence is modest (37%), but 3 of the 5 nearest reference specimens are *Royidris* ([see below](#)).

1.	Royidris myrmicinae	37.3%
2.	Nesomyrmex myrmicinae	9.5%
3.	Pheldole myrmicinae	6.8%

[See where this lands on the atlas →](#) [Where *Royidris* has been collected →](#)

[← Identify another specimen](#)

hf-hub:imageomics/bioclips-2 · probe 2026-08-29 · probabilities are the classifier's softmax over the 27 genera, not a calibrated chance of being right.

Most similar training specimens

cosine similarity of BioCLIP 2 embeddings; species labels are AntWeb's, not the model's



Royidris notorthotenes
myrmicinae
casent0002259
© April Nobile · [AntWeb](#) 0.950



Royidris robertsoni
myrmicinae
casent0077935
© April Nobile · [AntWeb](#) 0.901



Royidris notorthotenes
myrmicinae
casent0002257
© April Nobile · [AntWeb](#) 0.889



Monomorium bifidoclypeatum
myrmicinae
casent0173589
© April Nobile · [AntWeb](#) 0.861

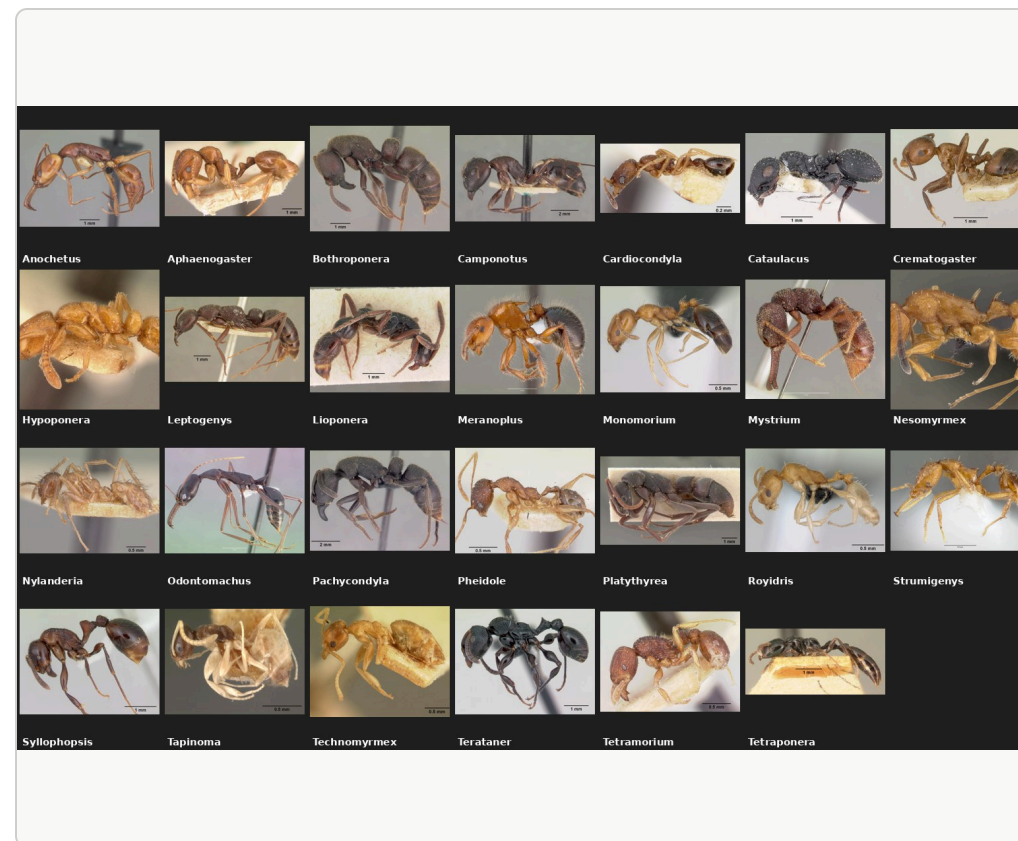


Nesomyrmex flavus
myrmicinae
casent0189276
© Erin Prado · [AntWeb](#) 0.858

Vitsika: *vitsika* is Malagasy for ant. Images © their photographers via [AntWeb](#) (CC BY-SA); metadata via GBIF (CC BY). [Methods & limitations](#)

Identification is the bottleneck, and it happens far from the field

- AntWeb holds **141,430** Malagasy ant records, **6,995** of them with images, across **60** imaged genera. Naming even the genus takes a specialist and a microscope.
- The specimens, the imaging and the determinations are curated and published by the California Academy of Sciences (San Francisco); the expertise sits outside Madagascar, and a field team waits for it.
- A photo-to-genus first pass that runs on a CPU box in Antananarivo in **~0.8 s** per image lets a local team triage, then send only the hard cases to a specialist.



One AntWeb profile shot per genus in scope (27). Sources: reports/01_explore.txt, README.md.

From GBIF to 1,236 clean images in 27 genera



Finding 1: GBIF's image cache is empty for AntWeb

4,215 of 4,354 cache requests returned 404 (96.8 %), uniformly across years and record age. antweb.org blocks scripts (Cloudflare) and datacenter IPs (403), including GBIF's own fetcher. The working source is the 2008-2014 bulk upload to Wikimedia Commons (33,081 files), matched by specimen code.

Finding 2: caste is not in GBIF's interpreted fields

GBIF maps sex to Male / Female / Other, so every worker arrives as null: 4,540 of 5,857 specimens. The caste has to be read from the verbatim record (dwc:sex): 4,539 workers, 673 queens, 640 males, and only workers are kept.

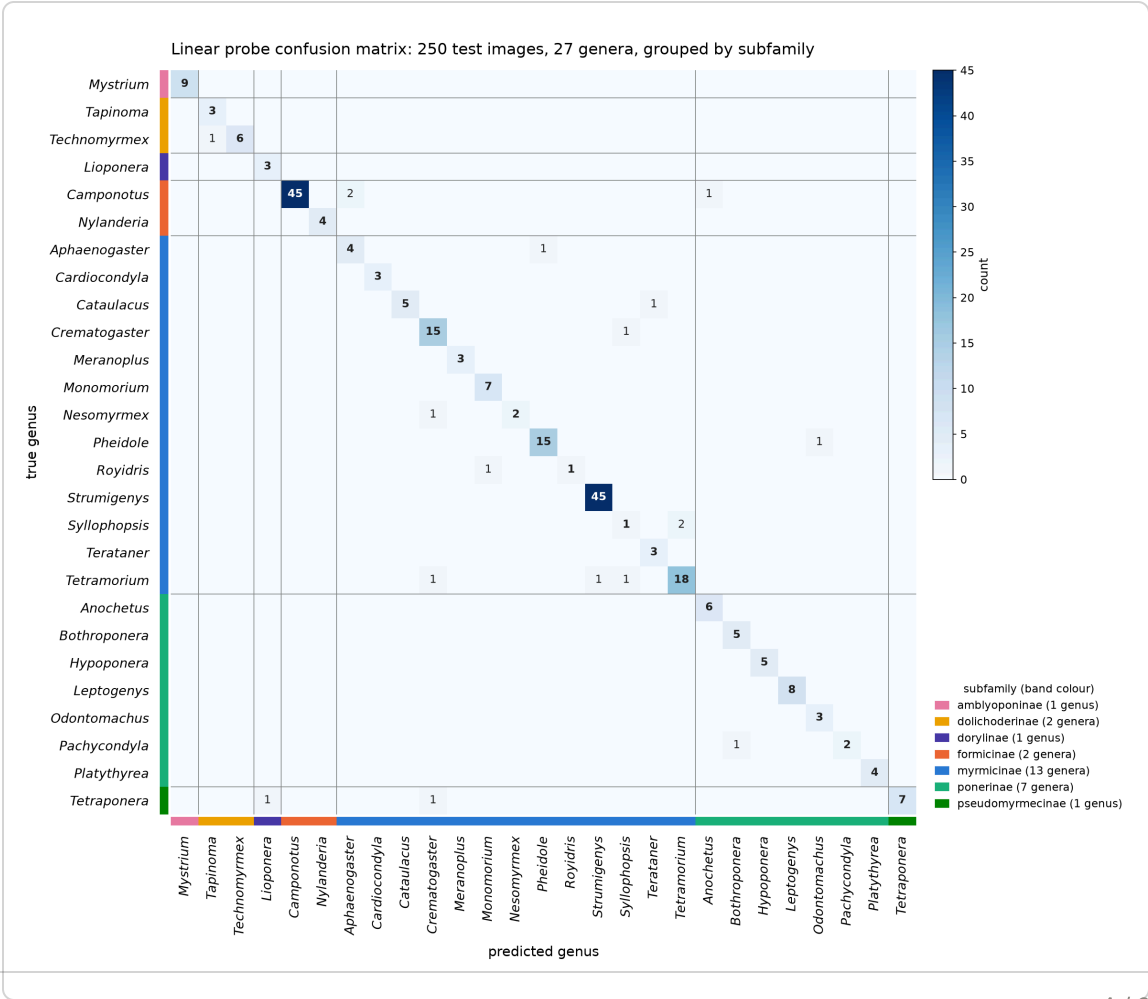
RESULTS

Linear probe on frozen BioCLIP 2 embeddings: 92.8 % top-1 on 250 held-out specimens

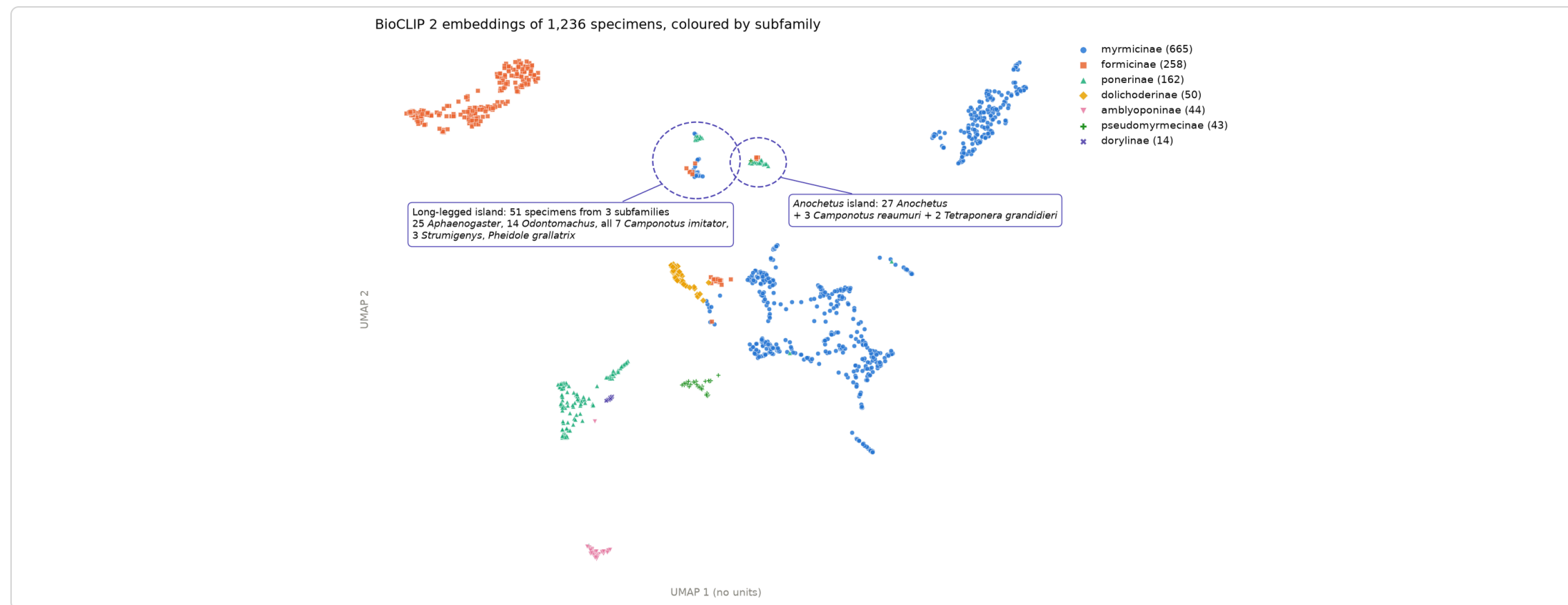
METHOD	TOP - 1	TOP - 3	MACRO - F1
Zero-shot, "a photo of {genus}, a genus of ant"	74.4 %	94.0 %	0.590
Linear probe (deployed)	92.8 %	98.4 %	0.885
Nearest neighbour, cosine	91.6 %	93.2 %	0.834

986 train / 250 test, 27 genera, split by genus (seed 42). Bare genus name as the zero-shot prompt: 32.8 % top-1. Embeddings: imageomics/bioclip-2 ViT-L/14, 768-d, no fine-tuning.

18 errors: 12 genuine look-alikes or species unseen in training (*Sylophopsis* vs *Tetramorium*, *Royidris* vs *Monomorium*), 3 poor photos, 2 odd angles, 1 unexplained. Almost every confusion stays inside a subfamily block.



The embedding space recapitulates taxonomy, with body-plan islands



Subfamilies form clean clusters; where the model "errs", it groups *Camponotus imitator* with long-legged *Aphaenogaster* and *Odontomachus*, a resemblance a taxonomist would recognise, not noise.

Three things this codebase already supports

1

Complete the image set

70.2 % of the manifest's images are missing. `05_download_antweb.py` + `dataset_full.csv` are ready for a residential IP or an AntWeb bulk export; re-thresholding brings *Vitsika*, *Tanipone*, *Carebara* and 9 more genera into scope (39 in total).

2

Honest generalisation and an open set

Group the split by species (one flag in `03_build_dataset.py`) and turn the nearest-neighbour similarity the API already returns into a "none of the 27" reject. Probabilities are already calibrated (temperature scaling on train folds, ECE 0.55 → 0.02).

3

Multi-view, multi-caste embeddings

The harvest already holds 6,634 head and 5,127 dorsal media rows plus 673 queens and 640 males; `config.yaml` (`views_priority`, `castes`) and `06_embed.py` make view-wise embedding and late fusion a configuration change.